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**Unlocking Grit: Exploring Predictors Among Education Students
in Purposive Communication Classes in the Philippines**

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- ☒ SSITRDC
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(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Teaching and Learning Initiatives		(8) Discipline/Field of Specialization Pedagogy

(9A) Abstract :

Grit, defined as perseverance and passion for long-term goals, is recognized as essential for academic success and personal development, but its specific predictors are not well-established. This descriptive-correlational study examined predictors of grit among 313 education students in one of the campuses of a state university in the Philippines during the second semester of the 2023-2024 academic year. Data were collected using the Social-Emotional Competence Questionnaire (SECQ), the Grit Scale, and the Academic Engagement Scale. Descriptive statistics, including percentages and means, were used for determining the socio-demographic characteristics. Pearson correlation coefficients assessed relationships between grit and continuous variables, while Eta correlations explored connections with categorical variables. Multiple regression analysis, including a stepwise method, identified significant predictors of grit. Results showed significant positive correlations between grit and socio-emotional competence, academic engagement, year level, digital resources, and financial support. The regression equation is: $Grit = 1.385 + 0.373 (Socio-Emotional Competence) + 0.220 (Academic Engagement) + 0.062 (Year Level) + 0.055 (Financial Support) - 0.046 (Internet Connectivity) - 0.262 (Civil Status)$. Socio-emotional competence and academic engagement are identified as the strongest positive predictors of grit, while internet connectivity and civil status negatively impact grit. These findings highlight the importance of developing

socio-emotional skills and academic engagement to enhance grit, offering valuable insights for educators aiming to improve student resilience and persistence.

(9B) Keywords: Academic engagement, grit, socio-emotional competence, predictors, regression analysis

(10) Introduction:

Grit, the combination of passion and perseverance for long-term goals, has emerged as a crucial predictor of academic success and personal achievement. According to Duckworth et al. (2007), grit involves two key dimensions: consistency of interest and perseverance of effort. These dimensions reflect a person's enthusiasm and sustained effort toward goals despite challenges and setbacks. While intelligence is a well-established factor in academic success, non-cognitive traits such as grit play a vital role in predicting students' persistence and achievement.

In education, various factors can predict grit, including socio-demographic characteristics, socio-emotional competence, and academic engagement. Socio-emotional learning (SEL) has gained attention for its impact on students' academic and personal lives, influencing grades, attitudes, emotions, and engagement (Zhou & Ee, 2012; Elias & Haynes, 2008; Durlak et al., 2011; Wang et al., 2016; Zins et al., 2007). Positive interpersonal relationships and emotional competence enhance enthusiasm for learning, contributing to sustainable success and self-confidence (Mercer & Dörnyei, 2020).

Influenced by personal characteristics, teaching methodologies, and the learning environment, academic engagement is another key predictor of grit. Engagement factors such as cognitive, metacognitive, affective, social, task-related, communicative, and language-related elements do not exist in isolation but are interconnected and mutually reinforcing. While much research has focused on the importance of academic engagement for improving educational quality and student success, the specific predictors of grit within foreign language learning contexts and their interplay with socio-demographic factors remain underexplored, highlighting the need for further investigation (Philp & Duchesne, 2016; Oga-Baldwin, 2019).

Despite the significance of these factors, more empirical research needs to be conducted to examine the predictors of grit among education students in a Philippine state university. This study aimed to address this gap by identifying the predictors of grit among education students enrolled in Purposive Communication classes. It focused on the influence of socio-demographic factors, socio-emotional competence, and academic engagement, providing insights into how these elements interact to shape students' grit and contribute to their educational success.

(11) Methodology

Research Design and Sampling

This descriptive-correlational study covered 313 education students of Capiz State University Pontevedra Campus during the second semester of School Year 2023-2024. These respondents were determined using G*Power software with medium effect size, 80% power, and 0.05 level of significance.

Instruments

The study data were collected using a Likert scale, with four separate sections for demographic information, grit, social-emotional competence, and academic engagement. These scales were

adopted from the *Social-Emotional Competence Questionnaire (SECQ)* of Zhou and Ee in 2012, the *Grit Scale* of Duckworth et al. in 2007, and the *Academic Engagement Scale* of Reeve in 2013). The questionnaires were face-validated by a panel of experts comprised of a psychometrician, an educational researcher, and an English language professor. The Social-Emotional Competence Questionnaire and Grit Scale comprised 25 and eight Likert items, respectively. Finally, the questionnaire on students' academic engagement presented 17 items that measure students' perspectives on their engagement toward academic affairs.

Data Gathering Procedure

Permission from the campus administrator and deans allowed the administration of questionnaires to the respondents. The questionnaires were in Google Docs format, and the link were sent to the respondents.

Data Analysis

Frequency and percentage were used to analyze descriptive data. In contrast, Eta correlation was used for a significant relationship between grit, socio-emotional competence, academic engagement, and socio-demographic characteristics. The Eta correlation coefficient is used when the independent variable is categorical, and the dependent variable is continuous.

The Pearson correlation coefficient was used to determine the significant relationship between grit and other variables. The Pearson correlation coefficient is applied when the independent and dependent variables are continuous. Multiple regression analysis using a step-wise method was used to determine the significant predictors of grit.

SPSS was used in processing the data.

(12) Results and Discussions

Socio-Demographic Characteristics of Education Students

The socio-demographic characteristics of the education students are shown in Table 1. Three hundred thirteen students served as respondents of this study.

Table 1. Socio-demographic characteristics of education students (N=313)

Variables	f	%
Sex		
Male	64	20.4
Female	249	79.6
Age		
19 years old and below	99	31.6
20 years old	81	25.9
21 years old	54	17.3
22 years old and above	79	25.2
Year Level		
Freshman	56	17.9
Sophomore	65	20.8
Junior	57	18.2

Senior	135	43.1
Parents Combined Family Income		
5,000 and below	132	42.2
5001-10,000	122	39.0
Above 10,000	59	18.8
Civil Status		
Single	305	97.4
Married, Live-in, Separated	8	2.6
Place of Origin		
Urban (City)	24	7.7
Town Proper	78	24.9
Rural-Upland	168	53.7
Rural-Coastal	43	13.7
Living Arrangements		
With family	289	92.3
Not with family	24	7.7
Financial Support		
Scholarships	99	31.6
Loans	23	7.3
Parental support	191	61.0
Support Systems		
Family	294	93.9
Peers, Mentors	19	6.1

Table 1. continued...

Variables	f	%
Motivational Factors		
Intrinsic motivation	24	7.7
Career aspirations	36	11.5
Personal goals	239	76.4
Others (Open communication)	14	4.5
Employment Status		
Full-time student	238	76.0
Full-time student with part-time job	56	17.9
Part-time student with part-time job	19	6.1

The socio-demographic characteristics of the students (N=313) in Table 1 are detailed as follows: The majority of the students are female, with 249 (79.6%) identifying as such, while 64 (20.4%) are male. In terms of age distribution, 99 students (31.6%) are 19 years old and below, 81 students (25.9%) are 20 years old, 54 students (17.3%) are 21 years old, and 79 students (25.2%) are 22 years old and above. The students' year levels vary, with 56 freshmen (17.9%), 65 sophomores (20.8%), 57 juniors (18.2%), and 135 seniors (43.1%).

When looking at their parents' combined family income, 132 students (42.2%) come from families earning 5,000 and below, 122 students (39.0%) from families earning between 5,001

and 10,000, and 59 students (18.8%) from families with income above 10,000. The civil status of the majority is single, with 305 students (97.4%) identifying as such, while eight students (2.6%) are married, live-in, or separated.

As to place of origin, 24 students (7.7%) come from urban cities, 78 (24.9%) from town proper areas, 168 (53.7%) from rural upland regions, and 43 (13.7%) from rural coastal areas. Most students live with their families (289 or 92.3%), whereas 24 students (7.7%) do not. Financial support is primarily provided by parents, with 191 students (61.0%) relying on parental support, 99 students (31.6%) benefiting from scholarships, and 23 students (7.3%) using loans.

Support systems are predominantly family-based, with 294 students (93.9%) depending on the family, while 19 (6.1%) rely on peers or mentors. Motivational factors vary, with 239 students (76.4%) driven by personal goals, 36 students (11.5%) by career aspirations, 24 students (7.7%) by intrinsic motivation, and 14 students (4.5%) by open communication with family and peers. Lastly, in terms of employment status, 238 students (76.0%) are full-time students, 56 students (17.9%) are full-time students with part-time jobs, and 19 students (6.1%) are part-time students with part-time jobs.

Gender differences in academic engagement and socio-emotional competence have been noted, with females often exhibiting higher levels of these traits (MacCann et al., 2020). As students advance in age and year level, their grit and academic performance levels tend to increase, likely due to accumulated experience and maturity (Datu et al., 2017). Family income significantly influences students' access to resources and academic motivation, highlighting the impact of socio-economic factors (Dumont et al., 2017).

Table 2 provides an overview of students' access to and utilization of educational resources. Access to educational materials is relatively balanced between books and libraries, used by 120 students (38.3%), and journals and online databases, accessed by 193 students (61.7%). In terms of technology access, smartphones are predominantly used by 255 students (81.5%), while 58 students (18.5%) utilize laptops or desktop computers.

Table 2. Access and utilization of educational resources among education students

Variables	f	%
Access to Educational Materials		
Book and Library	120	38.3
Journals, Online Database	193	61.7
Technology Access (Hardware)		
Smartphones	255	81.5
Laptops/ Desktop computers	58	18.5
Internet connectivity		
High-speed internet (Library)	50	16.0
Wi-Fi access (DICT)	149	47.6
Pisonet	70	22.4
Others (Mobile data)	44	14.1

Software and Applications		
Learning Management System	30	9.6
Educational Software	45	14.4
Online Collaboration (Zoom, Google Meet)	238	76.0
Digital Resources		
E-books	29	9.3
E-Journals	17	5.4
Online courses	93	29.7
Multimedia resources	106	33.9
Artificial intelligence tools	68	21.7

In terms of internet connectivity, 50 students (16.0%) have high-speed internet, 149 students (47.6%) rely on Wi-Fi access, 70 students (22.4%) use Pisonet, and 44 students (14.1%) access the internet through other means such as mobile data. The use of software and applications is also notable, with 30 students (9.6%) utilizing Learning Management Systems, 45 students (14.4%) using educational software, and 238 students (76.0%) employing online collaboration tools such as Zoom and Google Meet.

Digital resources are varied in their utilization: 29 students (9.3%) use e-books, 17 students (5.4%) access e-journals, 93 students (29.7%) engage with online courses, 106 students (33.9%) make use of multimedia resources, and 68 students (21.7%) employ artificial intelligence tools.

Access to digital resources and technology has been positively linked to increased student engagement and better academic performance, mainly when these resources facilitate interactive and personalized learning (Chen et al., 2020). Internet connectivity and digital tools are crucial for enhancing socio-emotional learning through collaborative platforms (Xie et al., 2020).

Correlates of Grit among Education Students

Table 3 provides a detailed correlation analysis between various variables and students' grit levels, including the correlation coefficients and their respective p-values. Significant correlations, indicated by p-values less than 0.05, are marked with an asterisk (*). The analysis reveals several key findings.

Table 3. Correlates of grit among education students

Variables	Test statistics (r/η)	p-value
Sex ^η	0.030	0.595
Age ^η	0.121	0.208
Year Level ^η	0.18	0.013
Table 3. continued...		
Variables	Test statistics (r/η)	p-value
Parents' combined family income ^η	0.050	0.683
Civil Status ^η	0.080	0.158

Place of Origin [‡]	0.154	0.058
Living Arrangements [‡]	0.154	0.058
Financial Support [‡]	0.071	0.454
Support Systems [‡]	0.075	0.185
Motivational Factors [‡]	0.096	0.410
Employment Status [‡]	0.040	0.778
Access to Educational Materials [‡]	0.057	0.314
Technology Access (Hardware) [‡]	0.027	0.635
Internet Connectivity [‡]	0.180	0.017
Software and Applications [‡]	0.018	0.951
Digital Resources [†]	0.205	0.010
Socio-Emotional Competence [†]	0.538	0.000
Academic Engagement [†]	0.508	0.000

p<0.05, [‡]Eta Correlation, [†]Pearson Correlation Coefficient

Firstly, there is a significant positive correlation between year level and grit ($\eta = 0.185$, $p = 0.013$), suggesting that their grit levels tend to increase as students advance through their academic stages. Similarly, internet connectivity shows a significant positive correlation with grit ($\eta = 0.180$, $p = 0.017$), indicating that better access to the internet is associated with higher levels of grit among students. Furthermore, access to digital resources is positively correlated with grit ($\eta = 0.205$, $p = 0.010$), suggesting that greater availability of digital resources is linked to increased grit.

In addition, socio-emotional competence demonstrates a strong positive correlation with grit ($r = 0.538$, $p = 0.000$), revealing that students with higher socio-emotional competence tend to exhibit significantly higher grit levels. Academic engagement also shows a strong positive correlation with grit ($r = 0.508$, $p = 0.000$), implying that students who are more academically engaged generally display higher grit levels.

Conversely, several variables, including sex, age, parents' combined family income, civil status, place of origin, living arrangements, financial support, support systems, motivational factors, employment status, access to educational materials, technology access (hardware), and software and applications, did not show significant correlations with grit ($p > 0.05$). These findings reflect the importance of academic engagement, socio-emotional competence, access to digital resources, and internet connectivity in fostering grit among students.

Grit is positively associated with socio-emotional competence and academic engagement, indicating that emotionally intelligent students are more persistent in their studies (Crede et al., 2017). Access to digital resources supports the development of grit by providing students with tools and opportunities for self-directed learning (Wang et al., 2019).

Predictors of Grit Among Education Students

Table 4 presents a multiple regression analysis to identify the predictors of grit among students. The table includes the unstandardized regression coefficients (B), t-values, and p-values for each independent variable.

Table 4. Predictors of grit among education students

Independent Variables	B	t	p-value
(Constant)	1.385	5.711	.000
Socio-Emotional Competence	.373	6.717	.000
Academic Engagement	.220	4.848	.000
Year Level	.062	3.661	.000
Financial Support	.055	2.536	.012
Internet connectivity	-.046	-2.133	.034
Civil Status	-.262	-2.130	.034

Note: R=.631, R-squared=.398 , F=33.655 , p<0.001

Socio-emotional competence and academic engagement are strong predictors of grit, with high levels of these traits leading to increased resilience and persistence in students (Muenks et al., 2017). The relationship between internet connectivity and grit is complex, as excessive reliance on digital resources may sometimes hinder the development of perseverance (Chang et al., 2018).

Regression Model and Summary

The regression model shows a multiple correlation coefficient (R) of 0.631, indicating a good level of student grit prediction. The R-squared value of 0.398 suggests that the independent variables included in the model explain approximately 39.8% of the variance in grit. The overall model is statistically significant, as indicated by the F-statistic of 33.655 ($p < 0.001$), confirming that the predictors reliably estimate grit.

The multiple regression model developed to predict grit among students includes socio-emotional competence, academic engagement, year level, financial support, internet connectivity, and civil status as independent variables. The regression equation is formulated as follows:

$$\text{Grit} = 1.385 + 0.373 (\text{Socio-Emotional Competence}) + 0.220 (\text{Academic Engagement}) + 0.062 (\text{Year Level}) + 0.055 (\text{Financial Support}) - 0.046 (\text{Internet Connectivity}) - 0.262 (\text{Civil Status})$$

The regression analysis identifies several significant predictors of grit among students. Socio-emotional competence emerged as a strong positive predictor, with a coefficient (B) of 0.373 ($t = 6.717$, $p < 0.001$), indicating that students with higher socio-emotional competence exhibit higher levels of grit. Similarly, academic engagement is a significant positive predictor ($B = 0.220$, $t = 4.848$, $p < 0.001$), suggesting that students who are more engaged academically tend to have higher grit levels. The year level of students also significantly predicts grit ($B = 0.062$, $t = 3.661$, $p < 0.001$), with students in higher academic years showing greater grit.

Financial support shows a positive relationship with grit ($B = 0.055$, $t = 2.536$, $p = 0.012$), implying that students receiving financial support are more likely to have higher grit.

Interestingly, internet connectivity is a significant negative predictor of grit ($B = -0.046$, $t = -2.133$, $p = 0.034$), suggesting that better internet access may be associated with lower grit levels. Civil status also negatively predicts grit ($B = -0.262$, $t = -2.130$, $p = 0.034$), with married, live-in, or separated students exhibiting lower grit than their single counterparts.

(13) Conclusions and Recommendations

Conclusions

Based on the findings of the study, the following conclusions were drawn:

The findings highlight the importance of socio-emotional competence, academic engagement, year level, financial support, internet connectivity, and civil status in predicting student grit. Socio-emotional competence and academic engagement are the strongest positive predictors, emphasizing their critical role in promoting resilience and perseverance. The unexpected negative relationship between internet connectivity and grit suggests a need for further investigation into how digital access impacts students' grit. These predictors collectively offer valuable insights for educational institutions aiming to cultivate grit in their student populations.

Recommendations

Given the findings of the study and conclusions made, the following recommendations were presented:

- **Enhance Socio-Emotional Learning:** Educators and educational policymakers may develop culturally tailored programs to build students' emotional resilience and social skills.
- **Promote Academic Engagement:** Educators may use real-world, relevant projects to deepen students' connection to their studies.
- **Provide Targeted Support Across Year Levels:** The student services arm of the university may offer stage-specific support to address students' evolving academic and career needs.
- **Address Financial Constraints:** The government may consider revisiting the sound expansion of financial aid and partnerships to reduce economic barriers to student success.
- **Investigate and Address Digital Access Issues:** The DICT and private telecommunications companies may improve digital access and affordability to support consistent academic engagement.
- **Support Students Based on Civil Status:** The guidance office and the educators may consider providing flexible resources and counseling for students with diverse family responsibilities.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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